UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence — Fall 2025
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Lecture 3: Stein's Method

Content:

1. Probability Metrics
2. Stein's Method

2.1 Probability Metrics

Definition 2.1.1 For any two random variables X, Y or their corresponding probability measures μ_X, μ_Y, the Kolmogorov–Smirnov distance between X and Y is defined to be

$$\begin{aligned} d_{KS}(X,Y) &:= \sup_{t\in\mathbb{R}} |\mu_X((-\infty,t]) - \mu_Y((-\infty,t])| \\ &= \sup_{t\in\mathbb{R}} |\mathbb{P}(X \le t) - \mathbb{P}(Y \le t)|. \end{aligned}$$

Remark:

- By definition, $X_n \xrightarrow{d} X$ if $d_{KS}(X_n, X) \to 0$. At the same time, convergence in Kolmogorov-Smirnov distance is stronger than convergence in distribution.[1]
- Note that $(-\infty, t] \in \mathcal{B}(\mathbb{R})$, the KS Distance can be bounded by the total variation:

$$\begin{aligned} d_{KS}(P,Q) &= \sup_{A: A=(-\infty,t]} |\mu_X(A) - \mu_Y(A)| \\ &\le \sup_{A\in\mathcal{B}(\mathbb{R})} |\mu_X(A) - \mu_Y(A)| = d_{TV}(X,Y) \end{aligned}$$

[1] It is also known that convergence of distribution is metrizable by Lévy–Prokhorov metric.

Definition 2.1.2 For any two random variables X, Y, the Wasserstein Distance between X and Y is defined to be

$$d_W(X,Y) = \sup_{f: f \text{ is 1-Lipschitz}} |\mathbb{E}f(X) - \mathbb{E}f(Y)|$$

here we called f is 1-Lipschitz if $|f(x) - f(y)| \leq |x - y|$ for any x, y.

Lemma 2.1.3 For any two random variables W, Z, if Z has density bounded by c w.r.t. the Lebesgue measure, then

$$d_{KS}(W,Z) \leq \sqrt{2c \cdot d_W(W,Z)}$$

Proof: Denote $f_x(z) = \mathbf{1}\{z \leq x\}$, we have $\mathbb{E}[f_x(W)] = \mathbb{P}(W \leq x)$, thus

$$d_{KS}(W,Z) = \sup_{x \in \mathbb{R}} \big|\mathbb{E}[f_x(W)] - \mathbb{E}[f_x(Z)]\big|$$

For any $\varepsilon > 0$, define $f_x^\varepsilon(z)$ as follow:

$$f_x^\varepsilon(z) := \begin{cases} 1, & \text{if } z \in (-\infty, x] \\ 1 - (z-x)/\varepsilon, & \text{if } z \in [x, x+\varepsilon] \\ 0, & \text{if } z \in [x+\varepsilon, +\infty) \end{cases}$$

Since f_x^ε is ε^{-1}-Lipschitz, $\varepsilon f_x^\varepsilon$ is 1-Lipschitz, by definition we have

$$\varepsilon \cdot |\mathbb{E}[f_x^\varepsilon(W)] - \mathbb{E}[f_x^\varepsilon(Z)]| \leq d_W(W,Z)$$

note that $f_x^\varepsilon \geq f_x$ and $f_Z(z) \leq c$, we have

$$\begin{aligned} \mathbb{E}[f_x(W)] - \mathbb{E}[f_x(Z)] &\leq \mathbb{E}[f_x^\varepsilon(W)] - \mathbb{E}[f_x(Z)] \\ &\leq (\mathbb{E}[f_x^\varepsilon(W)] - \mathbb{E}[f_x^\varepsilon(Z)]) + \mathbb{E}[f_x^\varepsilon(Z) - f_x(Z)] \\ &\leq \frac{d_W(W,Z)}{\varepsilon} + \int_x^{x+\varepsilon} \left(1 - \frac{z-x}{\varepsilon}\right) f_Z(z) dz \\ &\leq \frac{d_W(W,Z)}{\varepsilon} + \int_x^{x+\varepsilon} \left(1 - \frac{z-x}{\varepsilon}\right) c \cdot dz \\ &= \frac{d_W(W,Z)}{\varepsilon} + \frac{c\varepsilon}{2} = \sqrt{2c \cdot d_W(W,Z)} \end{aligned}$$

here as the last inequality holds for all $\varepsilon > 0$, taking the minimum over all $\varepsilon > 0$ leads to the final result. For the other side, consider defining $\overline{f_x^\varepsilon} \leq f$:

$$\overline{f_x^\varepsilon}(z) := \begin{cases} 1, & \text{if } z \in (-\infty, x-\varepsilon] \\ (x-z)/\varepsilon, & \text{if } z \in [x-\varepsilon, x] \\ 0, & \text{if } z \in [x, +\infty) \end{cases}$$

2.2 Stein's Method

In this section, we use W for arbitrary distributions; $Z \sim \mathcal{N}(0,1)$ to denote the standard Gaussian distribution, and $\mathcal{A}$ as the Stein's operator s.t.

$$(\mathcal{A}f)(x) = f'(x) - xf(x).$$

For clarity, we also omit $\sqrt{2\pi}$ factors from the calculations below.

Lemma 2.2.1 (Stein's lemma) If $Z \sim N(0,1)$, then for any absolutely continuous $f : \mathbb{R} \to \mathbb{R}$ with $\mathbb{E}|f'(Z)| < \infty$, we have $\mathbb{E}[f'(Z)] = \mathbb{E}[Zf(Z)]$.

Proof: Suppose f has compact support on (a,b), then

$$\begin{aligned}\sqrt{2\pi} \cdot \mathbb{E}[f'(Z)] &= \int_a^b e^{-x^2/2} f'(x)dx = \left[f(x)e^{-x^2/2}\right]_a^b - \int_a^b (-x)f(x)e^{-x^2/2}dx \\ &= \int_a^b xf(x)e^{-x^2/2}dx = \sqrt{2\pi} \cdot \mathbb{E}[Zf(Z)]\end{aligned}$$

For the general case, we first claim that $\mathbb{E}|Zf(Z)| < \infty$, and $\mathbb{E}|f(Z)| < \infty$. WLOG, assume $f(x) = 0$ if $x \in (-\infty, 0]$, then

$$\begin{aligned}\mathbb{E}|Zf(Z)| &\propto \int_0^\infty |tf(t)|e^{-t^2/2}dt = \int_0^\infty t\left|\int_0^t f'(x)dx\right| e^{-t^2/2}dt \\ &\leq \int_0^\infty \int_0^t t|f'(x)|e^{-t^2/2}dxdt = \int_0^\infty |f'(x)| \left(\int_x^\infty te^{-t^2/2}dt\right)dx \\ &= \int_0^\infty |f'(x)|e^{-x^2/2}dx = \mathbb{E}|f'(Z)| < \infty\end{aligned}$$

As a result, $\mathbb{E}|f(Z)| \leq \sup_{x\in[0,1]} |f(x)| + \mathbb{E}|Zf(Z)| < \infty$.

Given $\mathbb{E}|f(Z)|,\ \mathbb{E}|Zf(Z)|,\ \mathbb{E}|f'(Z)|$ are finite, define $f_n(t) = f(t)g_n(t)$, where

$$g_n(t) := \begin{cases} 1, & \text{if } |t| \in [0,n] \\ 2 - |x|/n, & \text{if } |t| \in [n,2n] \\ 0, & \text{if } |t| \in [2n, +\infty) \end{cases}$$

Note that $|g_n|,\ |g_n'| \leq 1$, and $f_n \to f$ pointwise, we have

$$|tf_n(t)| \leq |tf(t)|,\ |f_n'| = |f'g_n + fg_n'| \leq |f'| + |f|$$

By DCT, $\mathbb{E}[Zf_n(Z)] \to \mathbb{E}[Zf_n(Z)]$, $\mathbb{E}[f_n'(Z)] \to \mathbb{E}[f'(Z)]$, while f_n is a.c. with compact support, we know $\mathbb{E}[f_n'(Z)] = \mathbb{E}[Zf_n(Z)]$, thus

$$\mathbb{E}[Zf(Z)] = \lim_{n\to\infty} \mathbb{E}[Zf_n(Z)] = \lim_{n\to\infty} \mathbb{E}[f_n'(Z)] = \mathbb{E}[f'(Z)]$$

Lemma 2.2.2 (Stein's lemma) Given f, define the Stein operator

$$(\mathcal{A}f)(x) = f'(x) - xf(x). \tag{1}$$

1. If $W \sim \mathcal{N}(0,1)$, then $\mathbb{E}[(\mathcal{A}f)(W)] = 0$ for all absolutely continuous f with $\mathbb{E}|f'(W)| < \infty$.
2. If W satisfies that $\mathbb{E}[(\mathcal{A}f)(W)] = 0$ for all absolutely continuous f with $\mathbb{E}|f'(W)| < \infty$, then $W \sim \mathcal{N}(0,1)$.

Proof: We only prove the second one. If we can show that the following differential equation has an "adequate" solution:

$$f'(w) - wf(w) = \mathbf{1}\{w \leq x\} - \Phi(x)$$

then plugging it into the condition we can immediately imply that $W \sim \mathcal{N}(0,1)$:

$$0 = \mathbb{E}[(\mathcal{A}f)(W)] = \mathbb{P}(W \leq x) - \Phi(x)$$

To solve this differential equation, note that

$$\frac{d}{dw}\left(e^{-w^2/2}f(w)\right) = e^{-w^2/2}(f'(w) - wf(w)) = e^{-w^2/2}(\mathbf{1}\{w \leq x\} - \Phi(x))$$

which implies

$$e^{-w^2/2}f(w) = \int_w^\infty e^{-t^2/2}(\Phi(x) - \mathbf{1}\{t \leq x\})dt + C$$

$$f(w) = e^{w^2/2}\int_w^\infty e^{-t^2/2}(\Phi(x) - \mathbf{1}\{t \leq x\})dt + Ce^{w^2/2}$$

To make f adequate, let $C = 0$, then f' is bounded and f remains to be absolutely continuous.

3 Informal Derivation of Stein's Equation

Stein's equation is not only valid for Gaussian distribution; for every distribution, it is possible to formulate a 'Stein's lemma' analogous to Lemma 2.2.2. That is, for any distribution $\mathcal{D}$, it is possible to find a Stein operation $\mathcal{A}_\mathcal{D}f$ such that

$$W \sim \mathcal{D} \Leftrightarrow \mathbb{E}[\mathcal{A}_\mathcal{D}f(W)] = 0$$

for every absolutely continuous $f : \mathbb{R} \to \mathbb{R}$ with bounds on its derivatives.

First approach Let W has density ϕ supported on $\mathbb{R}$. Note that

$$\mathbb{E}[f(W)] = \int f(x)\phi(x)\ dx = \underbrace{\int f(x+t)\phi(x+t)\ dx}_{:=h(t)},$$

for any $t \in \mathbb{R}$. Note that the left hand side is independent of t, as such, $h(t)$ is a constant function. Thus,

$$h'(0) = 0 \Rightarrow \int (f'(x)\phi(x) + f(x)\phi'(x))\ dx = 0.$$

Namely,

$$\mathbb{E}\left[f'(W) + f(W)\frac{\phi'(W)}{\phi(W)}\right] = 0. \tag{2}$$

Plugging $\phi(x) = e^{-x^2/2}$ immediately yields (1). If the density have discontinuities, additional care is necessary. For instance, suppose the density is supported on $[a, \infty)$. Then, (2) modifies to

$$\mathbb{E}\left[f'(W) + f(W)\frac{\phi'(W)}{\phi(W)} + f(a)\phi(a)\right] = 0.$$

Second approach Note that

$$\mathbb{E}[f(W)] = \int f(x)\phi(x)\ dx = \underbrace{\int f(xt)\phi(xt)t\ dx}_{:=\bar{h}(t)},$$

Once again, $\bar{h}$ is constant in t. As such,

$$\bar{h}'(1) = 0,$$

which again gives a Stein equation.

Remark. *These approaches will potentially give two different Stein equations, both of which are perfectly valid. In fact, a Stein operator (associated with a distribution) need not be unique.*

Exercise 1. Using the machinery above, identify a Stein's equation for exponential distribution, $\mathrm{Exp}(1)$, and prove that it indeed characterizes $\mathrm{Exp}(1)$.